

Master Prompt — Full-Stack Kisan Vyapar Portal Generation

> Copy and paste this entire prompt into your AI IDE. It is structured in rigorous sections so the AI treats each domain area with equal priority and builds a complete, production-grade application without placeholders.

THE PROMPT

You are building a production-grade, fully functional agricultural marketplace and farmer portal called **Kisan Vyapar Portal**. The tech stack is strictly: **Next.js 14 (App Router)**, **TypeScript**, **Tailwind CSS**, **Framer Motion**, **Supabase (Auth + Database + Storage)**, and **Razorpay (Cards/UPI/NetBanking) + Cash on Delivery (COD)** for payments. The project must be deployable to **Vercel** with zero configuration issues.

This is NOT a template or mockup. Every feature must be wired to Supabase, fully functional with transactional integrity, server-side role-based access control (RBAC), and persistent database records.

SECTION 1 — UI DESIGN SYSTEM (HIGHEST PRIORITY)

The UI must look like it was designed by a top-tier digital product agency (think: Apple Store minimalism, Aesop typography & organic elegance, Linear dashboard clarity, and Shopify Dawn storefront structure). It must feel modern, fresh, calm, and trustworthy—never cluttered, cartoonish, or stereotypical.

Follow these rules with zero exceptions:

Typography

- Use **Inter** or **Plus Jakarta Sans** for body text, navigation, and data elements.
- Use a refined, warm serif font (**DM Serif Display** or **Playfair Display**) for hero headings, crop titles, and agricultural resource headers.
- Font sizes: Hero headings `56–72px`, Section headings `36–44px`, Crop/Product titles `18–22px`, Body text `15–16px`, Captions/labels `12–13px` in uppercase tracked-out (letter-spacing: `0.1em`).
- Line height: `1.2` for headings, `1.6` for body text. Never use default line heights.

Color Palette (Botanical Agricultural System)

- **Primary background:** `#FAFCFA` (warm organic off-white).
- **Secondary background:** `#F3FAF4` (soft sage tint).
- **Primary text:** `#1E2A22` (deep forest charcoal).
- **Secondary / muted text:** `#617064` (slate moss).

- **Accent brand green:** #2D7A46 (rich agricultural emerald) — used for primary CTAs, active links, and status highlights.
- **Light accent:** #6FBF78 (fresh sprout green) — used for badges, subtle borders, and focus rings.
- **Warm harvest gold / warning:** #B7791F (used for pending states, fresh harvest tags, and seasonal ribbons).
- **Destructive / error:** #DC2626 . **Success:** #16A34A .
- All cards and containers must use background #FFFFFF with a subtle border (border: 1px solid rgba(0,0,0,0.06)) and a refined shadow (box-shadow: 0 1px 3px rgba(0,0,0,0.04), 0 4px 12px rgba(0,0,0,0.03)). Never use heavy, dark borders.

Spacing & Layout

- Use an 8px grid system. All padding, margins, and gaps must be multiples of 8 (8, 16, 24, 32, 48, 64, 80, 120).
- Maximum content width: 1440px, centered. Horizontal page padding: 64px on desktop, 24px on tablet, 16px on mobile.
- Section vertical padding: 80–120px on desktop, 40–56px on mobile. Never less than 64px between major storefront sections.
- Marketplace Crop Grid: 4 columns on desktop, 3 on small desktop, 2 on tablet, 1 or 2 on mobile (375px). Grid gap: 24px. No masonry layouts.

Micro-Interactions & Animations (Framer Motion)

Every interaction must feel tactile, smooth, and purposeful. Implement all of the following:

1. **Page transitions:** Wrap the storefront and dashboards in `AnimatePresence` . Pages fade in (`opacity: 0 → 1`) and slide up (`y: 20 → 0`) over 0.4s with ease `[0.25, 0.1, 0.25, 1]` .
2. **Crop card hover:** On hover, the crop image scales to `1.05` over 0.5s with `cubic-bezier(0.4, 0, 0.2, 1)` . The harvest freshness badge pulses gently. A "Quick View" button slides up from below the card (`y: 20 → 0, opacity: 0 → 1`) with a staggered 0.1s delay.
3. **Add-to-Cart / Reserve animation:** When a buyer clicks "Add to Cart", animate a ghost image of the crop (fixed position, cloned from the product image) flying in a parabolic bezier arc toward the cart icon in the header. The ghost scales from `1 → 0.2` , rotates slightly (`10deg`), and fades out. Simultaneously, the cart icon triggers a spring bounce (`scale: 1 → 1.35 → 1` , spring stiffness: 400), and a small "+1" badge pops in. Orchestrate with Framer Motion's `useAnimation` .
4. **Button interactions:** All buttons scale to `0.97` on press (`whileTap`). Primary CTAs feature a subtle 0.2s emerald shimmer sweep animation on hover using background-position shift on a linear gradient.
5. **Scroll-triggered reveals:** Every section, crop card, community card, and resource block animates into view using `whileInView` (`opacity: 0 → 1, y: 30 → 0` over 0.6s) with `staggerChildren: 0.08` for grids.
6. **Cart & Weather slide-over drawers:** Open from the right as slide-over panels (`x: 100% → 0`) with a backdrop blur overlay (`backdrop-blur-sm`). Items inside stagger in (`staggerChildren: 0.05`). Removing an item animates it out (`opacity: 0, height: 0, x: 50`) before removal.

7. **Crop Image gallery on PDP:** Main crop photo crossfades smoothly on thumbnail selection (`AnimatePresence mode="wait"`). Thumbnail click triggers a subtle scale pulse. Implement pinch-to-zoom on mobile.
8. **Skeleton loading states:** Every dynamic query (crops, orders, weather, community) must display animated shimmer skeleton placeholders using CSS background-position gradients. Never show blank white space or layout shift (CLS = 0).
9. **Toast notifications:** Slide in from the top-right, auto-dismissing after 3s with a shrinking emerald progress bar at the bottom. Entrance: slide from right + fade. Exit: slide to right + fade.
10. **Sticky Header & Role Switcher:** Sticky header shrinks smoothly (80px → 64px) on scroll with glassmorphism backdrop blur. Mobile menu expands as a full-screen drawer with staggered links animating in from the left.

Component Quality Rules

- All images must use `next/image` with explicit aspect ratios, priority loading for above-the-fold assets, and blur placeholders.
- Accessible focus-visible rings: `2px offset ring in #2D7A46` for keyboard navigation.
- Form inputs must use animated floating labels (labels animate up on focus, not plain placeholder text).
- Full mobile responsiveness tested at 375px, 768px, 1024px, and 1440px. No action should ever require mouse hover.

SECTION 2 — STOREFRONT, MARKETPLACE & PORTAL PAGES

First-Open Role Selection Modal / Screen

On first visit, present a clean, welcoming role selector before registration:

- **Heading:** "How will you participate in Kisan Vyapar Portal?"
- **Subheading:** "Direct farm-to-buyer agricultural marketplace connecting producers directly with wholesale and retail buyers."
- Two equal-weight interactive cards:
 - "I am a Farmer" (List crops, receive verified orders, set minimum quantities, manage harvest schedule, check local weather, ask community questions).
 - "I am a Buyer" (Browse verified crops, direct farm pricing, place reservations/orders, track shipments via COD/Razorpay).
- Include "Browse Marketplace as Guest" and "Sign In to Existing Account".
- Preserve the selected role through registration and store it in the user's Supabase profile.

Homepage

- **Hero Section:** Full-width authentic farm lifestyle imagery with high-contrast text overlay. Animated headline with word-by-word reveal: "Connecting Indian Farmers Directly to Wholesale & Retail Buyers."

Search bar with category selector and location detector. Auto-rotating hero carousel with progress indicators.

- **Role Chooser Banner:** Quick-toggle callout cards directing users to "Start Selling Crops" or "Buy Fresh Farm Produce".
- **Category Grid:** Instant shortcut cards with photos and icons: *Grains & Cereals, Pulses & Legumes, Fresh Fruits, Vegetables, Spices, Oilseeds, and Organic Produce.*
- **Freshly Harvested Crops:** 4-column crop grid filtered by `harvest_date` descending, showcasing live "Harvested Today" or "Harvested 2 Days Ago" ribbons.
- **Weather & Mandi Advisory Teaser:** Compact dashboard card displaying local temperature, rainfall probability, humidity, and a general agricultural advisory note (with location permission prompt).
- **Agricultural Resource Hub Highlights:** Carousel of featured guides on Soil Health (Loamy vs. Sandy soil), Plant Disease Prevention, and Integrated Pest Management.
- **Trust & Verification Badges:** Infinite horizontal ticker showing: *100% Direct Trade, Government-ID Verified Farmers, Transparent Fair Pricing, Strict Quality Grading, Secure Razorpay & COD Payments.*
- **Testimonials:** Verified farmer and buyer stories stored in and rendered from Supabase.
- **Footer:** Multi-column footer with brand logo, dynamic category links, APMC/Mandi info links, weather advisory disclaimer, terms, privacy, and emergency contact.

Marketplace / Crop Listing Page (PLP)

- **Filters Sidebar (Desktop) / Bottom Sheet (Mobile):** Category selector, Price Range Slider (dual-thumb in INR ₹/kg or ₹/quintal), Quantity & Unit filter (kg, Quintal, Tonne, Crate, Box), Quality Grade filter (Grade A / Export Quality, Grade B / Standard, Grade C), Harvest Freshness filter (< 24 hrs, < 3 days, < 7 days, Upcoming Harvest), Location filter (State and District dropdowns), Delivery Type (Farm Pickup available vs Seller Door Delivery), and Farmer Verification toggle. All query parameters sync with URL.
- **Sort Dropdown:** Price Low-to-High, Price High-to-Low, Harvest Date (Freshest First), Highest Stock, Highest Farmer Rating.
- **Active Filter Chips:** Interactive tags with one-click remove (×) animation.
- **Quick View Modal:** Centered modal showing crop photos, variety, grade, minimum order quantity, harvest date, farm district, and an instant "Add to Cart" button.

Crop Detail Page (PDP)

- **Breadcrumb Navigation:** `Marketplace > Category > Crop Name (Variety)`.
- **Image Gallery:** Large primary image with thumbnail selector below. Zoom on desktop hover; mobile pinch-to-zoom. Support for up to 5 images per crop.
- **Crop Information:** Title in serif font, Price display in INR (e.g., `₹2,800 / Quintal` with ₹/kg breakdown), Harvest Freshness Ribbon ("*Harvested 2 days ago*"), Minimum Order Quantity (MOQ) indicator, Available Stock vs Reserved Stock bar.
- **Farmer Trust Strip:** Farmer display name & Farm name, District, State (full address hidden until order acceptance for privacy), Verified Farmer Badge with verification date, completed platform orders count & buyer satisfaction rating.

- **Fulfillment Selector:** Radio options for "Farm Pickup" or "Farmer Delivery" (with auto-calculated transport fee).
- **Quantity Selector:** Increment/decrement buttons locked to integer units, respecting Minimum Order Quantity and maximum available stock.
- **Stock-Aware CTA:** "Add to Cart" button with flying image animation; disabled with "Out of Stock" or "Sold Out" state when stock is 0.
- **Detailed Accordions:** Crop & Variety Specifications, Harvest & Storage, Cultivation & Soil Type, Farmer Terms & Pickup Instructions, and Buyer Reviews & Ratings.
- **Report Listing Button:** Modal to report fraudulent claims or incorrect crop listings.
- **Related Crops:** Horizontal carousel of similar crops from the same agricultural district.

Cart Page & Slide-Over Drawer

- Slide-over drawer on header icon click + dedicated `/cart` page.
- Itemized crop details: Crop image, title, variety, unit price, quantity adjuster, weight total (in kg/quintals), and line item total in INR.
- Automatic constraint validation: Prevents buyers from purchasing below the farmer's MOQ or exceeding available stock.
- Delivery Fee & Platform Fee calculation breakdown.
- Coupon code input with validation against Supabase `coupons` table.
- Subtotal, delivery surcharge, tax/mandi cess (if applicable), and Grand Total in INR.
- "Proceed to Secure Checkout" button with shimmer animation.

Multi-Step Checkout Page

- **Step 1: Delivery or Pickup Selection:** Select saved delivery address or view farm pickup address.
- **Step 2: Buyer Contact Information:** Full name, verified mobile phone number for delivery coordination, email.
- **Step 3: Payment Selection:**
 - **Cash on Delivery (COD):** Pay farmer/driver upon physical delivery/pickup inspection.
 - **Online Payment via Razorpay:** UPI, Google Pay, PhonePe, Cards, NetBanking.
- **Step 4: Order Review:** Itemized crop snapshot, weight summary, seller terms acknowledgement.
- **Place Order CTA:** Triggers server-side atomic stock reservation. On Razorpay, loads the Razorpay checkout overlay; on COD, marks order as "Pending Farmer Acceptance" and redirects to the confirmation page.

Order Confirmation & Live Tracking Page

- Animated SVG path draw checkmark on completion.
- Order Reference Number: `ORD-KVP-10001`.
- Farm-to-Buyer Order Timeline with visual state markers:
 1. *Order Placed / Payment Verified*

2. *Accepted by Farmer*

3. *Harvested / Packed*

4. *Dispatched / Ready for Pickup*

5. *Delivered / Order Completed*

- Download printable invoice/delivery receipt.
- "Continue Shopping" and "Message Support" CTAs.

Farmer Dashboard & Workspace (/farmer)

- **Overview Dashboard:** Metric cards: Active Listings, Total Sales (₹), Pending Order Requests, Low Stock Alerts. Recent Order Requests table with one-click "Accept" or "Reject" (with reason). Quick action CTA: "+ List New Crop".
- **Crop Listing Wizard (/farmer/listings/new):** Multi-step form with save-as-draft capability:
 - Step 1: Crop Name, Category, Variety, Quality Grade (Grade A, B, C).
 - Step 2: Total Quantity, Unit (kg, Quintal, Tonne, Crate), Minimum Order Quantity, Price per Unit (₹).
 - Step 3: Harvest Date, Availability Window, Delivery/Pickup capabilities.
 - Step 4: Soil type & farming method details (Loamy/Sandy/Black soil, Organic/Traditional).
 - Step 5: Drag-and-drop crop images (up to 5 images uploaded to Supabase Storage bucket `crop-images`).
- **Listing Management Table:** Columns: Image, Crop Title, Category, Stock, Price/Unit, Status (*Draft, Pending Approval, Active, Paused, Sold Out*), Actions (Edit, Pause, Delete).
- **Farmer Orders Management (/farmer/orders):** Filterable by: New Requests, Accepted, In-Transit/Dispatched, Completed, Cancelled. Action buttons: "Accept Order", "Mark as Packed", "Dispatch Order" (with transport notes & tracking phone).

Buyer Dashboard (/account)

- Active Orders & History with tracking timeline.
- Saved Delivery Addresses book (CRUD with default address toggle).
- Saved Wishlist / Favorite Farms.
- Profile Settings: Name, email, phone number, language preference.
- Post-purchase review form: Rate farmer and crop quality with 1–5 stars and text feedback.

Weather Hub (/weather)

- City, District, or PIN code search + one-click browser geolocation button.
- Clean dashboard displaying: Current Temperature (°C), Humidity %, Wind Speed (km/h), Precipitation Probability %, and 5-Day Agricultural Forecast.
- **Agricultural Decision Card:** Practical planting, spraying, and harvesting advice tailored to the weather forecast (clearly marked as general guidance).

Agricultural Resource Hub (/resources)

- Searchable and filterable knowledge base categorized by:
 - *Soil Types & Health*: Detailed guides for Loamy, Sandy, Clayey, and Black soils.
 - *Plant Diseases & Prevention*: Symptoms, prevention, and organic remediation.
 - *Pest Management*: Integrated Pest Management, biological controls.
 - *Sustainable Practices*: Crop rotation, water conservation, drip irrigation, cover crops.
- Article detail pages with reading time, high-resolution imagery, and actionable agronomic takeaways.

Farmer Community Forum (/community)

- Interactive discussion forum for farmers and buyers.
- Topic tags: *Crop Diseases*, *Mandi Prices & Trends*, *Equipment & Tech*, *Organic Farming*, *Government Schemes*.
- Create discussion post with title, topic tag, rich text body, and optional image upload (Supabase Storage).
- Reply threads, upvoting, and built-in "Report" flag to prevent spam.
- User privacy: Displays only author display name and broad district; never displays email or personal phone.

SECTION 3 — ADMIN & MODERATION PANEL (/admin)

The admin console must use a completely separate layout (no public storefront header or footer). It must be strictly protected by Supabase Auth with server-side role verification (`profiles.role === 'admin'`). Non-admin users attempting access must be redirected to `/`.

Admin UI Design

- Sleek collapsible sidebar with **Lucide icons**.
- Color scheme: Pure white sidebar (`#FFFFFF`), light gray dashboard content (`#F8F9FA`), and emerald primary accent (`#2D7A46`).
- All tables must support search, column sorting, status filtering, and pagination (20 items per page).
- Destructive actions (suspending a user, rejecting a listing, deleting content) must require a confirmation modal.
- Charts built with **Recharts** matching the brand palette.

Admin Dashboard (/admin)

- **Key Metrics**: Total Gross Merchandise Value (GMV in ₹), Total Completed Orders, Active Farmers count, Active Buyers count, Crops Listed, Pending Listing Approvals.
- **Revenue & Orders Trend**: Line/Area chart displaying sales volume over the last 7, 30, and 90 days.
- **Recent Orders Queue**: Top 10 most recent transactions with status indicators.

- **Flagged Items Widget:** Quick alerts for reported community posts and disputed orders.

Admin Management Screens

- **Crop Listing Moderation (/admin/listings):** Full table of all submitted crops. Detail drawer: Review farmer identity, crop photos, price/unit, harvest date, and claims. Actions: Approve Listing, Reject Listing (with rejection reason sent to farmer), or Request Modification.
- **Farmer Verification & User Management (/admin/users):** Master list of all users filtered by role. Farmer Verification Queue: Review farmer land details, farm location, or credentials. One-click toggle for Verified Farmer badge. Suspend / Reactivate account with audit notes.
- **Order & Dispute Oversight (/admin/orders):** Platform-wide order tracking table. Filter by payment status (COD, Paid via Razorpay, Refunded) and fulfillment status. Admin support override to cancel an order, trigger Razorpay refund, or resolve delivery disputes.
- **Community & Content Moderation (/admin/community):** Queue of reported forum questions and replies. Actions: Dismiss report, Hide post, Delete content, or Temporarily mute abusive users.
- **Resource Hub CMS (/admin/resources):** CRUD interface for agricultural guides, soil articles, and plant disease advisories. Markdown editor with live preview and image uploader.
- **Categories & Market Settings (/admin/settings):** Add/edit crop categories and standard units. Configure platform parameters: Minimum order rules, announcement banner text, support contact info. Searchable Audit Log table.

SECTION 4 — SUPABASE DATABASE SCHEMA

Create the following tables in Supabase with proper relationships, Foreign Keys, Row Level Security (RLS) policies, and performance indexes:

1. profiles

```
id (uuid, PK, ref auth.users.id cascade), email (text, unique), full_name (text), phone (text),
role (enum: 'farmer', 'buyer', 'admin', default 'buyer'), avatar_url (text), district (text),
state (text), pincode (text), created_at (timestampz), updated_at (timestampz)
```

2. farmer_profiles

```
id (uuid, PK), user_id (uuid, unique, ref profiles.id cascade), farm_name (text),
farm_description (text), farm_address (text), crops_grown (text[]), is_verified (boolean,
default false), verification_notes (text), created_at (timestampz)
```

3. categories

```
id (uuid, PK), name (text, unique), slug (text, unique), description (text), image_url (text),
sort_order (int), created_at (timestampz)
```

4. crop_listings

id (uuid, PK), farmer_id (uuid, ref profiles.id cascade), category_id (uuid, ref categories.id), title (text), slug (text, unique), variety (text), grade (enum: 'Grade A', 'Grade B', 'Grade C', 'Organic Certified'), description (text), unit (enum: 'kg', 'quintal', 'tonne', 'crate', 'bag'), price_per_unit (numeric(10,2)), min_order_quantity (int, default 1), stock_quantity (int), reserved_quantity (int, default 0), harvest_date (date), available_from (date), available_until (date), district (text), state (text), pickup_available (boolean), delivery_available (boolean), delivery_fee_per_unit (numeric(10,2)), status (enum: 'draft', 'pending_approval', 'active', 'paused', 'sold_out', 'rejected'), rejection_reason (text), created_at, updated_at

5. crop_images

id (uuid, PK), listing_id (uuid, ref crop_listings.id cascade), image_url (text), sort_order (int), alt_text (text)

6. addresses

id (uuid, PK), user_id (uuid, ref profiles.id cascade), full_name (text), phone (text), address_line1 (text), address_line2 (text), landmark (text), city (text), district (text), state (text), pincode (text), is_default (boolean, default false), created_at

7. cart_items

id (uuid, PK), buyer_id (uuid, ref profiles.id cascade), listing_id (uuid, ref crop_listings.id cascade), quantity (int), delivery_choice (enum: 'pickup', 'delivery'), created_at

8. orders

id (uuid, PK), order_number (text, unique, e.g. 'ORD-KVP-10001'), buyer_id (uuid, ref profiles.id), farmer_id (uuid, ref profiles.id), subtotal (numeric(10,2)), delivery_fee (numeric(10,2)), platform_fee (numeric(10,2)), discount_amount (numeric(10,2)), total_amount (numeric(10,2)), payment_method (enum: 'cod', 'razorpay'), payment_status (enum: 'pending', 'paid', 'failed', 'refunded'), razorpay_order_id (text), razorpay_payment_id (text), fulfillment_status (enum: 'pending', 'accepted', 'packed', 'dispatched', 'delivered', 'cancelled', 'disputed'), shipping_address (jsonb), delivery_type (text), notes (text), placed_at, accepted_at, dispatched_at, delivered_at, cancelled_at

9. order_items

id (uuid, PK), order_id (uuid, ref orders.id cascade), listing_id (uuid, ref crop_listings.id set null), crop_title (text), variety (text), unit (text), quantity (int), unit_price (numeric(10,2)), line_total (numeric(10,2))

10. order_timeline

id (uuid, PK), order_id (uuid, ref orders.id cascade), status (text), note (text), created_by (uuid, ref profiles.id), created_at

11. reviews

id (uuid, PK), order_id (uuid, ref orders.id), listing_id (uuid, ref crop_listings.id), buyer_id (uuid, ref profiles.id), farmer_id (uuid, ref profiles.id), rating (int check 1..5), review_text (text), created_at

12. resources

id (uuid, PK), title (text), slug (text, unique), topic (enum: 'soil_types', 'plant_diseases', 'pest_management', 'crop_management', 'sustainable_farming'), summary (text), content (text), thumbnail_url (text), read_time (text), is_published (boolean, default true), created_at

13. community_posts & 14. community_replies

posts: id, author_id, topic, title, body, image_url, status ('active', 'flagged', 'removed'), upvotes, created_at, updated_at
replies: id, post_id, author_id, body, status ('active', 'flagged', 'removed'), created_at

15. content_reports & 16. coupons & 17. site_settings & 18. audit_logs

reports: id, reporter_id, entity_type ('listing', 'post', 'reply'), entity_id, reason, status ('pending', 'reviewed', 'dismissed')
coupons: id, code (unique), discount_percent, discount_fixed, min_order_amount, times_used, max_uses, is_active, expires_at
settings: site_name, tagline, contact_email, phone, currency_code ('INR'), currency_symbol ('₹'), announcement_text
audit_logs: id, actor_id, action, entity_name, entity_id, details (jsonb), created_at

Row Level Security (RLS) Policies

- **profiles:** Public read for display name/avatar; users can update own profile; admins full access.
- **crop_listings:** Public read for `status = 'active'`. Farmers can view, create, edit own listings. Admins moderate all.
- **orders:** Buyers read own orders; farmers read incoming orders for their crops; admins read all.
- **cart_items:** Authenticated buyers can only read and mutate their own cart.
- **community_posts & replies:** Public read when `status = 'active'`; authenticated write; authors can edit/delete; admins moderate.
- **resources:** Public read for `is_published = true`; admin write only.

Storage Buckets (Supabase Storage)

- `"crop-images"` — public read; authenticated farmers can upload.
- `"farmer-documents"` — authenticated read/write for own documents; admin read.
- `"community-uploads"` — public read; authenticated users can upload.
- `"site-assets"` — public read; admin write.

Database Functions & Triggers:

- **Order Number Sequence:** Trigger on `orders` insert generating sequential numbers: `'ORD-KVP-' || LPAD(nextval('order_seq')::text, 5, '0')`.

- **Automatic updated_at:** Trigger updating `updated_at = now()` on row updates.
 - **Stock Reservation:** PostgreSQL transaction function decrementing `stock_quantity` and incrementing `reserved_quantity` atomically upon order creation.
-

SECTION 5 — TECHNICAL REQUIREMENTS

Project Structure (Next.js 14 App Router):

```
/app
  /(marketplace)
    /layout.tsx          ← Storefront header, role status, footer
    /page.tsx            ← Homepage with role selector, fresh crops, weather widget
    /crops/page.tsx      ← Crop Listing Page (PLP) with filters & sorting
    /crops/[slug]/page.tsx ← Crop Detail Page (PDP)
    /cart/page.tsx       ← Shopping cart page
    /checkout/page.tsx   ← Multi-step checkout (COD & Razorpay)
    /orders/[orderNumber]/page.tsx ← Order confirmation & tracking timeline
    /weather/page.tsx    ← Weather forecast & farm planning hub
    /resources/page.tsx  ← Agricultural knowledge hub
    /resources/[slug]/page.tsx ← Resource guide detail page
    /community/page.tsx  ← Community forum & questions
    /community/[id]/page.tsx ← Discussion thread & replies
    /about, /contact, /policies...
  /(farmer)
    /farmer/layout.tsx    ← Farmer portal layout with dedicated nav
    /farmer/dashboard/page.tsx ← Farmer overview & metrics
    /farmer/listings/page.tsx ← Active & draft crop listings
    /farmer/listings/new/page.tsx ← Multi-step crop listing form
    /farmer/orders/page.tsx ← Incoming order requests & status updates
  /(buyer)
    /account/layout.tsx   ← Buyer account navigation
    /account/orders/page.tsx ← Buyer order history & detail drawer
    /account/addresses/page.tsx ← Saved address management
  /(admin)
    /admin/layout.tsx     ← Admin sidebar & header
    /admin/page.tsx       ← Master analytics dashboard
    /admin/listings/page.tsx ← Listing moderation queue
    /admin/users/page.tsx  ← Farmer verification & user controls
    /admin/orders/page.tsx ← Platform-wide orders & disputes
    /admin/community/page.tsx ← Flagged content moderation
    /admin/resources/page.tsx ← Resource hub CMS
    /admin/settings/page.tsx ← Site configuration & audit logs
  /api
    /razorpay/order/route.ts ← Create Razorpay payment order
    /webhooks/razorpay/route.ts ← Razorpay signature verification & status update
    /weather/route.ts       ← Server-side cached OpenWeatherMap adapter
  /components
    /ui                    ← Button, Input, Modal, Toast, Skeleton, Badge, etc.
    /marketplace           ← CropCard, FreshnessBadge, FiltersSidebar, CartDrawer, etc.
    /farmer                ← ListingForm, OrderStatusDropdown, StockBadge, etc.
    /admin                 ← AdminSidebar, MetricsCard, DataTable, ModerationModal, etc.
  /lib
    /supabase/client.ts    ← Browser Supabase client
    /supabase/server.ts    ← Server component Supabase client (SSR cookies)
    /razorpay.ts           ← Razorpay server SDK setup
    /utils.ts              ← formatINR, calculateFreshness, slugify
  /hooks
    ← useCart, useRole, useWeather, useDebounce
  /types
    ← TypeScript interfaces (Crop, Order, FarmerProfile, etc.)
  /public
    ← Static SVGs, logos, fallback placeholders
```

State Management:

- **Cart State:** React Context with `localStorage` persistence providing: `items`, `addItem`, `removeItem`, `updateQuantity`, `clearCart`, `subtotal`, `itemCount`.
- **Role State:** Stored in React Context and persisted to `localStorage` + Supabase user profile.
- **Auth State:** Supabase Auth listener integrated into a global React Context provider.

Performance & Security:

- Next.js Dynamic Imports (`next/dynamic`) for heavy components (Recharts, rich text editors).
- Incremental Static Regeneration (ISR) on crop pages (`revalidate: 60`) and resource guides (`revalidate: 3600`).
- Server-side rate limiting and input validation with **Zod** on all mutating API routes.
- Stored money amounts formatted in INR using 2-decimal numeric types (never floating-point inaccuracies).
- Webhook signature validation (`crypto.createHmac('sha256')`) for all Razorpay payment notifications.
- Total first-load JavaScript bundle kept strictly under 200KB.

Deployment:

- Fully deployable to **Vercel** with zero configuration.
- `.env.example` file must document all variables:

```
NEXT_PUBLIC_SUPABASE_URL=your_supabase_project_url
NEXT_PUBLIC_SUPABASE_ANON_KEY=your_supabase_anon_key
SUPABASE_SERVICE_ROLE_KEY=your_supabase_service_role_key
NEXT_PUBLIC_RAZORPAY_KEY_ID=your_razorpay_key_id
RAZORPAY_KEY_SECRET=your_razorpay_key_secret
RAZORPAY_WEBHOOK_SECRET=your_razorpay_webhook_secret
WEATHER_API_KEY=your_weather_api_key
NEXT_PUBLIC_SITE_URL=http://localhost:3000
```

- Complete `README.md` containing local setup instructions, database migration commands, and deployment checklist.

SECTION 6 — FIRST IMPLEMENTATION STEPS

Generate the project sequentially in this exact order:

1. Initialize the Next.js 14 App Router project with TypeScript, Tailwind CSS, and Framer Motion.
2. Configure Supabase clients (`client.ts` and `server.ts` with `@supabase/ssr`).
3. Create the complete PostgreSQL migration file for Supabase with all 18 tables, triggers, and RLS policies.
4. Build the core UI design system components (`Button`, `Input`, `Card`, `Badge`, `Modal`, `Toast`, `Skeleton`) with all specified micro-interactions.

5. Build the primary application layout: Header with Role Chooser and animated CartDrawer, Navigation, and Footer.
6. Build the Homepage with the Hero Carousel, Category shortcuts, Freshly Harvested grid, and Weather advisory teaser.
7. Build the Marketplace Crop Listing Page (PLP) with dynamic multi-filter sidebar and sort controls.
8. Build the Crop Detail Page (PDP) with the multi-image gallery, freshness ribbon, farmer trust strip, and animated Add-to-Cart button.
9. Implement Cart and Checkout flows with COD support and Razorpay payment integration.
10. Build the Order Confirmation and Live Order Tracking Timeline pages.
11. Build the Farmer Portal (`/farmer`): Dashboard, Multi-Step Crop Listing Wizard, and Order Fulfillment Queue.
12. Build the Buyer Account (`/account`): Order History, Address Book, and Crop Reviews.
13. Build the Weather Hub (`/weather`) with the server-side OpenWeatherMap adapter and agricultural guidance notes.
14. Build the Agricultural Resource Hub (`/resources`) with Soil and Plant Disease guides.
15. Build the Farmer Community Forum (`/community`) with post creation and moderation flags.
16. Build the Admin Console (`/admin`): Dashboard metrics, Listing Moderation, Farmer Verification, and Dispute management.
17. Final quality polish: Loading skeletons, empty states, error boundaries, 404 page, and 100% mobile responsiveness.

Generate all code files with complete, working implementations. Do not use placeholder comments like `// TODO` or `// implement later` . Every component, hook, and route handler must be fully written and ready to deploy.

BONUS: UI REFERENCE KEYWORDS

If the AI needs extra guidance on the visual mood and design finish, append this at the end of the prompt:

VISUAL REFERENCE KEYWORDS FOR UI QUALITY:

The Kisan Vyapar Portal should visually feel like an exquisite blend of:

- Apple Store's clean whitespace, layout restraint, and typography precision.
- Aesop's organic, calm warmth and editorial dignity.
- Linear's crisp, high-efficiency dashboard states and keyboard fluidity.
- Shopify Dawn theme's intuitive, robust e-commerce product flow.
- A modern botanical color palette: warm off-white (`#FAFCFA`), rich agricultural emerald (`#2D7A46`), soft sage tints (`#F3FAF4`), and harvest gold accents (`#B7791F`).

If in doubt about any design decision, choose the cleaner, more spacious, and more accessible option. Generous whitespace, readable typography, and tactile micro-animations are core features.